

Performance Comparison Report

Advanced Machine Learning for House Price Prediction

1. Problem Statement

The objective of this project was to develop and compare multiple machine learning models for predicting house prices. A baseline Linear Regression model and advanced machine learning algorithms, including Decision Tree, Random Forest, and Gradient Boosting, were trained and evaluated. Hyperparameter tuning was also performed to determine whether model performance could be improved.

2. Dataset Description

- Dataset: House Price Prediction Dataset
- Target Variable: Price
- Features:
 - Area
 - Bedrooms
 - Bathrooms
 - Stories
 - Parking
 - Furnishing status
 - Other housing characteristics

Since the dataset had been cleaned in previous tasks, it was preprocessed encoding categorical variables, scaling numerical features where necessary, and splitting the data into training and testing sets.

3. Models Used

- Linear Regression (Baseline)
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor
- Tuned Random Forest (GridSearchCV)

4. Evaluation Metrics

The following regression metrics were used:

- Mean Absolute Error (MAE)

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

5. Performance Comparison

Model	MAE	RMSE	R^2 Score
Linear Regression	970,043	1,324,507	0.6529
Decision Tree	1,195,266	1,625,670	0.4771
Random Forest	1,021,546	1,400,566	0.6119
Gradient Boosting	959,749	1,299,386	0.6660
Tuned Random Forest	1,035,775	1,421,223	0.6004

6. Key Findings

- Gradient Boosting produced the highest predictive performance.
- Linear Regression performed surprisingly well considering its simplicity.
- Decision Tree showed signs of overfitting and had the poorest performance.
- Random Forest improved over Decision Tree by reducing variance through ensemble learning.
- Hyperparameter tuning did not improve the Random Forest model, demonstrating that optimization does not always lead to better results.

7. Recommendations

- Additional feature engineering could further improve predictive performance.
- More extensive hyperparameter tuning using RandomizedSearchCV or a broader GridSearchCV parameter space could be explored.
- Advanced boosting libraries such as XGBoost, LightGBM, or CatBoost may provide even better results.

8. Conclusion

In this project, five regression models were developed and evaluated to predict house prices: Linear Regression, Decision Tree, Random Forest, Gradient Boosting, and a Tuned Random Forest. Model performance was assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 Score.

Among all the models, the **Gradient Boosting Regressor** achieved the best overall performance, recording the **lowest MAE (959,749)**, the **lowest RMSE (1,299,386)**, and the **highest R² Score (0.6660)**. These results indicate that it produced the most accurate predictions and explained approximately **66.6%** of the variation in house prices.

Although hyperparameter tuning was applied to the Random Forest model using **GridSearchCV**, the tuned model performed slightly worse than the original Random Forest. This demonstrates that hyperparameter tuning does not always improve model performance and that the effectiveness of tuning depends on selecting appropriate parameter ranges and the characteristics of the dataset.

Overall, **Gradient Boosting** was identified as the best-performing model because it learns iteratively by correcting the errors of previous trees, enabling it to capture complex patterns in the housing data more effectively than the other models. Therefore, it is the most suitable model for predicting house prices in this project.